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# Automatic analysis and sketch-based retrieval of architectural floor plans

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ABSTRACT

The contribution of this article is twofold. First, we propose a sketch-based system, namely the a.SCatch system, for querying a floor plan repository. Second, for generating such a repository, a novel complete system for floor plan analysis is presented. The latter system extracts the semantics from existing floor plans. We introduce novel preprocessing methods, e.g., the differentiation between thick, medium, and thin lines and the removal of components outside the convex hull of the outer walls. Especially, the use of the convex hull increases the performance of the final system. The a.SCatch system enables the user to easily access knowledge from past projects. The user searches for semantically similar floor plans just by drawing parts of the new plan. An algorithm extracts the semantic structure sketched by the architect on DFKI's *Touch* & *Write* table. Finally, the extracted structures are compared using the graph matching, and the most similar one is retrieved. In our experiments for floor plan analysis on a reference data set we compare our approach to other approaches available in the literature. We show that our floor plan analysis system outperforms previous systems. Also the performance of the sketch recognition system is quite high. Overall, the performance of the floor plan analysis, as well as of the retrieval, is already good for the use in practice.

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## 1. Introduction

During design process, architects use existing and already designed buildings as reference. These reference drawings are used to guide solutions for similar architectural situations. Whenever an architect has to solve a new architectural problem, his first task will be to search for similar projects. By studying one or several previous reference projects the architect tries to derive a solution for his current problem. This is a common approach in architecture, knowing about to use reference projects during the design process and the knowledge of reference projects is an essential skill of an architect.

These days electronic search in architecture is realized by searching for textual annotations. However describing architectural work just using textual information is not sufficient, as verbal descriptions are subjective and often imprecise. Thus a pure textual annotation of floor plans is too fuzzy for an efficient retrieval.

Hence, in order to support an architect in his early design phase, there is a need for a search tool which enables him or her to find similar projects. As already stated, a textual search approach is not sufficient and too fuzzy. Furthermore, as we are dealing with visual information it is more intuitive to formalize a query employing a visual query language. The language should be an abstraction of an original floor plan symbolism, in order to have an intuitive way for architects to formalize a search query.

Langenhan proposes a semantic structure to describe the content of a floor plan based on functional and spatial relations of different structural entities. This formalization results in a graph representation which can be used for the retrieval process. The a.SCatch system enables the user to easily access knowledge from past projects. The user searches for semantically similar floor plans just by drawing parts of the new plan.

However, it is a well known problem that it is difficult to generate the semantic database for already existing reference paper floor plans, also known as the bootstrapping problem. Therefore, our second major contribution is an automatic floor plan recognition system which analyzes the floor plans and finally retrieves the corresponding semantic information. Usually, floor plan analysis systems consist of information segmentation, followed by

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# 建筑平面图的自动分析与基于草图的检索

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摘 要

本文的贡献主要体现在两个方面。首先，我们提出了一种基于草图的系统——a.SCatch系统，用于查询楼层平面图知识库。其次，为构建此类知识库，我们还提出了一套全新的楼层平面图分析完整系统，该系统能够从现有楼层平面图中提取语义信息。我们引入了若干新颖的预处理方法，例如区分粗线、中线和细线，以及去除位于外墙凸包之外的构件。尤其值得注意的是，通过引入凸包概念，显著提升了最终系统的性能。a.SCatch系统使用户能够便捷地获取历史项目中的知识；用户只需绘制新方案的部分草图，即可检索语义上相似的楼层平面图。一种算法会从DFKI的Touch & Write交互桌上提取建筑师所绘制的语义结构，随后利用图匹配技术对提取的结构进行比对，并返回最相似的结果。在针对参考数据集的楼层平面图分析实验中，我们将所提方法与文献中现有的其他方法进行了比较，结果表明，我们的楼层平面图分析系统优于现有系统。此外，草图识别系统的性能亦相当优异。总体而言，无论是楼层平面图分析还是检索环节，其性能均已达到实际应用的要求。

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## 1. 引言

在设计过程中，建筑师以现有及已建成的建筑作为参考。这些参考图纸用于指导类似建筑情境下的解决方案。每当建筑师需要解决一个新的建筑问题时，其首要任务便是寻找相似的项目。通过研究一个或多个既有的参考项目，建筑师试图为当前问题提炼出相应的解决方案。这是一种在建筑设计中常见的方法；在设计过程中善于利用参考项目，并掌握相关知识，是建筑师的一项必备技能。

如今，建筑领域的电子检索主要通过搜索文本注释来实现。然而，仅以文本信息描述建筑作品是不充分的，因为文字描述具有主观性且往往不够精确。因此，对平面图进行纯文本标注过于模糊，难以实现高效的检索。

因此，为了在设计的早期阶段为建筑师提供支持，亟需一种能够帮助其检索相似项目的搜索工具。如前所述，基于文本的检索方法既不充分又过于模糊。此外，由于处理的是视觉信息，采用视觉查询语言来形式化查询更为直观。该语言应是对原始平面图符号体系的抽象，以便建筑师能够以直观的方式构建查询表达式。

Langenhan提出了一种语义结构，用于基于不同结构实体之间的功能与空间关系来描述平面图的内容。该形式化方法生成了一种可用于检索过程的图表示。a.SCatch系统使用户能够便捷地获取以往项目中的知识。用户只需绘制新方案的部分内容，即可搜索语义相似的平面图。

然而，一个众所周知的问题是，为已有的参考论文平面图生成语义数据库十分困难，这也被称为“自举问题”。因此，我们的第二项主要贡献是一个自动平面图识别系统，该系统对平面图进行分析，并最终检索出相应的语义信息。通常，平面图分析系统由信息分割步骤组成，随后

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structural analysis and finally, semantic information extraction and alignment. The retrieved structural and semantic information can be saved in a repository for later access during retrieval.

Note that this article is an extended version of the conference contribution by Weber et al. (2010), where the a.SCatch system was initially proposed. While Weber et al. (2010) provided only a short description of the retrieval system, this article presents more details on all involved processes. Furthermore, the floor plan recognition methods introduced by Ahmed et al. (2011a) are elaborated and evaluated in this paper as well. Finally, we propose novel post-processing techniques for the semantic floor plan analysis and report on results of the floor plan recognition as well as sketch recognition and floor plan retrieval.

The novel key techniques of our approach are: (i) the segmentation into thick, *medium*, and thin walls to improve preprocessing techniques; (ii) the extraction of the building boundary removing many false positives during text extraction; (iii) detection of wall edges using convex-concave hypothesis; (iv) use of patch based approach coupled with Speeded Up Robust Features (SURF) for finding doors locations in the given floor plan.

The rest of this paper is organized as follows. First, Section 2 gives an overview of the related work. Subsequently, Section 3 introduces the concepts of the a.SCatch system and discusses the details of the floor plan analysis system. Section 4 shows the performed experiments. Finally, Section 5 concludes the work.

## 2. Related work

This section provides an overview of related work to the different techniques used in this paper. Related work can be categorized into architectural background, sketch-based interfaces, symbol spotting, complete floor plan analysis systems, and graph matching.

### 2.1. Architectural background

Since the middle of the 1990s the approach of applying Case-Based Reasoning (CBR) to design and architectural tasks has been known as Case-Based Design (CBD). The case-base contains information on buildings that have already been built or designed, enabling the computer to adapt solutions accordingly, on its own or with help from the architects. Table 1 provides a brief overview of some CBD systems based on two studies published by Heylighen and Neuckermans (2001) and by Richter et al. (2007) regarding the proposed approach. The marked fields show whether the appropriate feature was realized in the concept.

The study by Richter et al. (2007) identifies an acquisition bottleneck in putting complete case descriptions (problem and solution) into the case-base. We assume this is due to a lack of adequate input strategies, indexing methods and knowledge management procedures. First of all, a user interface should support the graphical sketch-based workflow of architects combined with textual, schematic and tabular input strategies. Secondly, a light-weight indexing strategy is needed in contrast to the overall data storage method used. Thirdly, the problem and solution descriptions need to be stored according to the CBR paradigm. Most of the CBD prototypes do not properly implement this fundamental CBR attribute.

### 2.2. Sketch-based interfaces

Sketches are widely used in engineering and architectural fields as they are a familiar, efficient and natural way of expressing certain kinds of ideas. Feng et al. (2008), proposed an 2D dynamic programming approach for analyzing hand-drawn electronic circuits. Sezgin et al. (2001) introduced a system that combines multiple sources of knowledge to provide robust early processing for freehand sketching.

Sim-U-Sketch is a sketch-based interface for Simulink[1] (Kara and Stahovich, 2004) where users can construct functional Simulink models simply by drawing sketches on a computer screen. To support iterative design, Sim-U-Sketch allows users to interact with their sketches in real time to modify existing objects and add new ones.

The *COMIC* system (Os and Boves, 2003) is a large European project that studies multi-modal interactions in design applications using pen and speech. In multi-modal system methods, such as mode detection by Willems et al. (2005), are sufficient to improve the usability of these systems.

Spatial-Query-by-Sketch proposed by Egenhofer (1996) describes a visual spatial query language for geometric information systems. Yaner and Goel (2006) examines the retrieval and mapping tasks of visual analogy as constraint satisfaction problems.

### 2.3. Symbol spotting

A main issue in the floor plan analysis is symbol spotting. Therefore, we list some related work in symbol spotting in this section.

In the past, different pattern recognition techniques have been applied to symbol spotting. Belkasim et al. (1991), Li and Shen (1991), Adam et al. (2000) and Arajo and Kim (2007) used feature based description for the purpose of symbol spotting. Similarly symbol spotting based on structural representation of documents has been used by Lladoós et al. (2001) and Yan and Wenyin (2003). Furthermore, Tabbone et al. (2003) performed symbol spotting based on image segmentation. However, segmentation itself leads to errors, which are then propagated to the recognition.

Another idea to address symbol spotting is to use a vectorial image to spot the symbol rather than using a raster image. Messmer and Bunke (1996) and Rusiñol and Lladós (2006) used vectorized image for symbol spotting. In addition to vectorization, Rusiñol et al. (2010b)used indexing techniques to increase the efficiency of symbol spotting techniques. To further increase the scalability of the method Rusiñol et al. (2010a) used relational indexing of vectorial primitives. Dutta et al. (2011) proposed a method using hashing the shape descriptors of graph paths.

Nayef and Breuel (2010) used geometric primitives as feature points. These feature points are then searched using geometric matching algorithm. To further increase the performance as well as the accuracy of the method, Nayef and Breuel (2011) used statistical grouping for segmenting parts from line drawings. After grouping, symbol spotting is performed using the method by Nayef and Breuel (2010).

### 2.4. Floor plan analysis

In past, floor plan analysis has been performed for different purposes. Aoki et al. (1996) and Lladós et al. (1997) analyzed hand sketched floor plan and generated respective CAD representation. Whereas, Dosch and Masini (1999), Dosch et al. (2000), Lu et al. (2007) and Or et al. (2005) focused on analysis of 2D diagrams of floor plans so that their respective 3D model can be regenerated.

Wessel et al. (2008) proposed a method for extracting the room connectivity graphs from 3D architectural models. Based on this

[1] Simulink is an environment for multidomain simulation and Model-Based Design for dynamic and embedded systems. http://www.mathworks.com/products/simulink/: Last accessed 04/02/2010

结构分析，最后进行语义信息的提取与对齐。所检索到的结构信息和语义信息可存储于知识库中，以便在后续检索过程中调用。

请注意，本文是Weber等人（2010）在会议上发表的论文的扩展版本，该论文首次提出了a.SCatch系统。尽管Weber等人（2010）仅对检索系统进行了简要描述，但本文则详细阐述了所有相关流程。此外，本文还对Ahmed等人（2011a）提出的楼层平面识别方法进行了深入探讨与评估。最后，我们提出了用于语义楼层平面分析的新型后处理技术，并报告了楼层平面识别、草图识别以及楼层平面检索的相关实验结果。

本方法的创新性关键技术包括：(i) 将墙体划分为厚墙、中等厚度墙体和薄墙，以优化预处理流程；(ii) 提取建筑边界，在文本提取过程中有效去除大量误检；(iii) 基于凸凹性假设检测墙体边缘；(iv) 采用基于图像块的方法，并结合加速稳健特征（SURF）算法，用于在给定的平面图中定位门的位置。

本文的其余部分安排如下。首先，第2节概述相关研究工作；随后，第3节介绍a.SCatch系统的基本概念，并详细讨论楼层平面分析系统的实现细节；第4节展示所开展的实验；最后，第5节对全文进行总结。

## 2. 相关工作

本节概述了与本文所采用的各类技术相关的研究工作。相关研究可分为以下几类：体系结构背景、基于草图的交互界面、符号识别、完整的平面布局分析系统以及图匹配。

### 2.1. 架构背景

自20世纪90年代中期以来，将基于案例的推理（CBR）应用于设计与建筑任务的方法被称为基于案例的设计（CBD）。该案例库中存储了已建成或已设计建筑的相关信息，使计算机能够自主地或在建筑师的协助下，据此对解决方案进行相应的调整。表1根据Heylighen和Neuckermans（2001）以及Richter等人（2007）发表的两项研究，对所提出的若干CBD系统进行了简要概述。其中标注的栏目显示了相应功能是否已在概念层面得以实现。

Richter 等人（2007）的研究指出，在将完整的案例描述（问题与解决方案）录入案例库的过程中存在获取瓶颈。我们认为，这主要是由于缺乏适当的输入策略、索引方法以及知识管理流程所致。首先，用户界面应支持建筑师基于图形草图的工作流程，并辅以文本、示意图和表格等多种输入方式。其次，相较于整体的数据存储机制，还需采用一种轻量级的索引策略。最后，问题与解决方案的描述必须按照案例推理（CBR）范式进行存储；然而，大多数 CBD 原型并未正确实现这一 CBR 的基本属性。

### 2.2 基于草图的交互界面

草图在工程和建筑领域被广泛使用，因为它们是一种熟悉、高效且自然的表达方式。某些类型的想法。Feng 等人（2008）提出了一种用于分析手绘电子电路的二维动态规划方法。Sezgin 等人（2001）则介绍了一个融合多种知识源的系统，以实现对徒手草图的稳健早期处理。

Sim-U-Sketch 是一种基于草图的 Simulink 界面，由 Kara 和 Stahovich 于 2004 年提出，用户只需在计算机屏幕上绘制草图即可构建功能性的 Simulink 模型。为支持迭代式设计，Sim-U-Sketch 允许用户实时与草图交互，以修改现有对象并添加新对象。

COMIC系统（Os和Boves，2003）是一个大型欧洲项目，旨在研究在设计应用中基于笔式输入与语音的多模态交互。在多模态系统方法中，诸如Willems等人（2005）提出的模式检测技术，足以提升此类系统的可用性。

Egenhofer（1996）提出的基于草图的空间查询描述了一种用于几何信息系统的可视化空间查询语言。Yaner和Goel（2006）将视觉类比的检索与映射任务视为约束满足问题进行研究。

### 2.3. 符号检测

在平面图分析中，一个主要问题在于符号识别。因此，我们在本节中列举了一些与符号识别相关的研究工作。

过去，不同的模式识别技术已被应用于符号检测。Belkasim 等人（1991）、Li 和 Shen（1991）、Adam 等人（2000）以及 Arajo 和 Kim（2007）采用了基于特征描述的方法进行符号检测。同样地，Lladoós 等人（2001）和 Yan 与 Wenyin（2003）则利用文档的结构化表示来进行符号检测。此外，Tabbone 等人（2003）基于图像分割开展了符号检测。然而，分割过程本身会引入误差，并将这些误差传递至后续的识别环节。

另一种解决符号检测问题的思路是采用矢量图像而非栅格图像进行符号检测。Messmer和Bunke（1996）以及Rusiñol和Lladós（2006）均采用了矢量化图像来实现符号检测。除了矢量化之外，Rusiñol等人（2010b）还引入了索引技术，以提升符号检测方法的效率。为进一步增强该方法的可扩展性，Rusiñol等人（2010a）提出了基于矢量图元的关联索引方法。Dutta等人（2011）则提出了一种通过哈希图路径的形状描述子来实现符号检测的方法。

Nayef和Breuel（2010）采用几何基元作为特征点，并通过几何匹配算法对这些特征点进行搜索。为进一步提升该方法的性能与精度，Nayef和Breuel（2011）在从线画中进行部件分割时引入了统计分组技术。分组完成后，再采用Nayef和Breuel（2010）提出的方法进行符号检测。

### 2.4 地面平面分析

以往，平面图分析曾被用于多种目的。Aoki 等人（1996）和 Lladós 等人（1997）对手绘平面图进行了分析，并分别生成了相应的 CAD 表达；而 Dosch 和 Masini（1999）、Dosch 等人（2000）、Lu 等人（2007）以及 Or 等人（2005）则侧重于对平面图的二维图形进行分析，以便重建相应的三维模型。

Wessel 等人（2008）提出了一种从三维建筑模型中提取房间连通性图的方法。基于此，

[1] Simulink is an environment for multidomain simulation and Model-Based Design for dynamic and embedded systems. http://www.mathworks.com/products/simulink/: Last accessed 04/02/2010

**Table 1**
Overview Case-Based Design systems.

| | Storage | | | Input | | | Output | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Floor plans + text | Abstraction | Topology | Graphic | Verbal | Adaption | Reference projects | Applying solutions | Graphical Information | Learning | Subproblems | Semantic net | Analogy |
| *Archie-II* Domeshek et al. (1994) | X | X | | | X | | X | | X | | X | X | |
| *CADRE* Hua et al. (1996) | X | X | X | X | X | X | | X | X | X | X | | |
| *FABEL* Voss (1997) | X | X | X | X | X | X | X | X | X | | X | | X |
| *IDIOM* Lottaz et al. (1998) | | | X | X | X | X | | X | X | | | | |
| *PREC.* Oxman and Oxman (1993) | X | X | | | X | | X | | X | | X | X | |
| *SEED L.* Flemming (1994) | | X | | | X | X | X | X | X | | X | | |
| *SL_CB* Lee et al. (2002) | X | X | | | X | X | X | X | X | | | | |
| *TRACE* Mubarak (2004) | | X | X | X | X | | X | X | X | | | | |
| *CaseBook* Inanc (2000) | X | | X | X | X | | X | | | | | | |
| *MONEO* Taha (2006) | X | X | | | X | | X | | X | | | | X |
| *CBA* Lin and Chiu (2003) | X | X | | | X | | X | | | | | X | |
| *DYNAMO* Heylighen and Neuckermans (2001) | X | X | | X | X | | X | | X | X | | X | |

connectivity graph a fast and efficient shape retrieval from an architectural database can be achieved. Macé et al. (2010) proposed a method to detect rooms in the architectural floor plan images. The method is based on a recursive decomposition of the images until convex regions are found.

Heras et al. (2011) performed the segmentation of walls from architectural floor plans. The segmented walls can be used for different purposes like, 3D reconstruction or building boundary construction.

In this paper, a novel method for automatic analysis of floor plans is introduced. It uses similar ideas as those proposed by Macé et al. (2010) and furthermore introduces new ideas like wall edges extraction, boundary detection. This general method can also be used for 3D reconstruction and generating the CAD format. However, the current focus is to extract the structure and the semantic information of the floor plan.

### 2.5. Graph matching

As for the retrieval algorithm, subgraph matching approaches seem to be promising. Given the problem that a query graph is formalized at run time and a database of model graphs exists a priori, techniques dealing with these conditions are of interest. Furthermore, the algorithm has to handle the fact that there might not be an exact match in the database.

Graph matching is challenging in presence of large databases (Bunke, 1997, 2000). Consequently, methods for indexing and preprocessing are essential methods. The main idea of the graph filtering is to use simple features to reduce to number of feasible candidates. Another concept clustering is used for grouping similar graphs. In principle, given a similarity (or dissimilarity) measure, such as GED (Bunke and Shearer, 1998), clustering algorithms can be applied.

Messmer and Bunke (1999) proposed a decision tree approach for subgraph matching. They are using the permutated adjacency matrix from a graph to build a decision tree. This technique is quite efficient during run time, as a decision tree is generated beforehand which contains all model graphs. Unfortunately the method has to determine all permutations of the adjacency matrices. Thus, as discussed in their experiments, the method is practically limited to graphs with a maximum 19 vertices.

## 3. Concept of a.SCatch

Based on the results of Langenhan and Petzold (2010), the aim of a.SCatch is to implement a system which takes advantage of the semantical information. A semantic search will be realized by sketching a concept of an architectural problem and triggering a search for similar projects of the past. Therefore several subtasks have to be solved:

1. Automatic extraction of the semantic structure of older projects,
2. extraction of the semantic structure from the sketch of the architect,
3. retrieval of similar floor plans from the repository,
4. visualization of the results and the interaction with the user interface.

The following sections will discuss the further details of subtasks of the a.SCatch system. Note that we tried to avoid using many parameters but still few exists. These parameters are optimized on validation set which include 10 images from original floor plans dataset. The test set contains the remaining 80 images which are used for evaluation of the proposed method.

### 3.1. Automatic extraction of the semantic structure from floor plans

The input data of our system is available in binary format.[2] First, segmentation algorithms are applied to separate the various types of

[2] The actual image size is $2479 \times 3508$. For making the analysis process more efficient, isotropic down scaling to $1413 \times 2000$ has been applied.

表1 基于案例的设计系统概述。

| | Storage | | | Input | | | Output | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Floor plans + text | Abstraction | Topology | Graphic | Verbal | Adaption | Reference projects | Applying solutions | Graphical Information | Learning | Subproblems | Semantic net | Analogy |
| *Archie-II* Domeshek et al. (1994) | X | X | | | X | | X | | X | | X | X | |
| *CADRE* Hua et al. (1996) | X | X | X | X | X | X | | X | X | X | X | | |
| *FABEL* Voss (1997) | X | X | X | X | X | X | X | X | X | | X | | X |
| *IDIOM* Lottaz et al. (1998) | | | X | X | X | X | | X | X | | | | |
| *PREC.* Oxman and Oxman (1993) | X | X | | | X | | X | | X | | X | X | |
| *SEED L.* Flemming (1994) | | X | | | X | X | X | X | X | | X | | |
| *SL_CB* Lee et al. (2002) | X | X | | | X | X | X | X | X | | | | |
| *TRACE* Mubarak (2004) | | X | X | X | X | | X | X | X | | | | |
| *CaseBook* Inanc (2000) | X | | X | X | X | | X | | | | | X | |
| *MONEO* Taha (2006) | X | X | | | X | | X | | X | | | | X |
| *CBA* Lin and Chiu (2003) | X | X | | | X | | X | | | | | X | |
| *DYNAMO* Heylighen and Neuckermans (2001) | X | X | | X | X | | X | | X | X | | X | |

通过连通性图，可以从建筑数据库中实现快速且高效的形状检索。Macé等人（2010）提出了一种用于检测建筑平面图中房间的方法，该方法基于对图像的递归分解，直至识别出凸区域。

Heras 等人（2011）对建筑平面图中的墙体进行了分割。分割得到的墙体可用于多种用途，例如三维重建或建筑物边界构建。

本文提出了一种用于平面图自动分析的新方法。该方法借鉴了Macé等人（2010）所提出的思路，并进一步引入了墙体边缘提取、边界检测等新概念。这一通用方法还可用于三维重建及生成CAD格式文件，但当前的重点在于提取平面图的结构信息与语义信息。

### 2.5. 图匹配

就检索算法而言，子图匹配方法似乎颇具前景。鉴于查询图在运行时被形式化，而模型图数据库则预先存在这一问题，针对此类条件的技术研究具有重要意义。此外，该算法还需处理数据库中可能不存在精确匹配的情况。

在大型数据库环境下，图匹配问题具有较高的难度（Bunke，1997，2000）。因此，索引与预处理方法至关重要。图过滤的核心思想是利用简单特征来减少可行候选集的规模。此外，聚类技术可用于对相似图进行分组。原则上，给定一种相似性（或相异性）度量，例如GED（Bunke和Shearer，1998），即可应用各类聚类算法。

Messmer和Bunke（1999）提出了一种用于子图匹配的决策树方法。他们利用图的置换邻接矩阵来构建决策树。该技术在运行时效率较高，因为决策树是在……之前生成的。

该方法包含所有模型图。然而，该方法必须枚举邻接矩阵的所有排列，因此，正如其在实验中所讨论的那样，该方法在实践中仅适用于最多包含19个顶点的图。

## 3. a.Catch的概念

基于Langenhan和Petzold（2010）的研究结果，a.SCatch的目标是构建一个能够充分利用语义信息的系统。通过勾勒出某一建筑问题的概念，并触发对以往类似项目的检索，从而实现语义搜索。为此，需要解决若干子任务：

1. 老旧项目语义结构的自动提取，2. 建筑师草图中语义结构的提取，3. 从知识库中检索相似的平面布局，4. 结果的可视化及与用户界面的交互。

以下各节将详细讨论a.SCatch系统各项子任务的具体内容。需要注意的是，我们尽量避免使用过多参数，但仍存在少量参数。这些参数是在包含原始平面图数据集中的10张图像的验证集上进行优化的。测试集则由剩余的80张图像组成，用于评估所提出方法的性能。

### 3.1. 从平面图中自动提取语义结构

本系统的输入数据以二进制格式提供。[2] 首先，采用分割算法对不同类型的

[2] The actual image size is 2479 × 3508. For making the analysis process more efficient, isotropic down scaling to 1413 × 2000 has been applied.

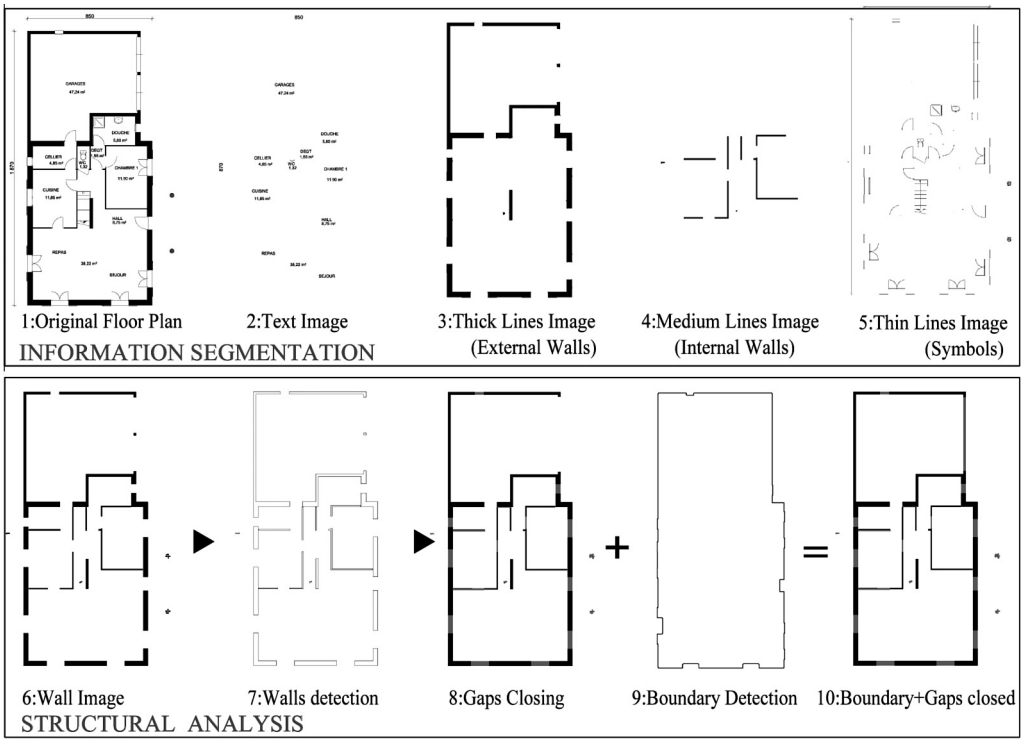



**Fig. 1.** Information Segmentation and Structural Analysis.

information from one another (see Section 3.1.1). Second, the structure of the extracted information is analyzed to retrieve the structure of the rooms (see Section 3.1.2). Finally, a semantic analysis is applied to retrieve the functions of the rooms, respectively (see Section 3.1.3).

### 3.1.1. Information segmentation

Floor plans contain information that collectively help an architect to express the actual dynamics of the building. During floor plan analysis, different types of information need to be interpreted at different points of time. Based on the divide and conquer strategy a process of information segmentation is performed. One of the key points of the proposed method is its fine segmentation of different types of information available in floor plans, e.g., walls, symbols, text etc. This is required because information, which is not needed for a specific step, is just noise and might lead to incorrect results.

First, text/graphics segmentation is performed using the methods presented by Ahmed et al. (2011b). This method is based on method by Tombre et al. (2002) with number of improvements specifically for the floor plans. Text/graphics segmentation separate the text from the graphics in the floor plan image. Text image (see Fig. 1.2) is later used in semantic analysis for extraction of room functions.

The graphics image is further segmented into walls and other building elements like door, windows, etc. Dosch et al. (2000) and Macé et al. (2010) used a thick/thin line separation algorithm for the separation of walls from the symbols. This algorithm separates the image into two images, i.e., a *thick lines image* containing the walls (both external and internal walls) and a *thin lines image* containing the symbols. We have enhanced this method by adding a third kind of lines, i.e., *medium lines*. In the proposed method, thick (external walls), medium (internal walls), and thin (symbols) lines image can be achieved by sequentially performing erosions followed by dilations. To get thick lines image, it is performed $n$ times to remove everything but the thick walls, where $n$ is computed empirically based on thickness of external walls from training set.

The thick line image is then subtracted from graphics image, to obtain an intermediate image which contains medium and thin lines. On the resulting image, it is performed only one time to get medium lines image. Finally, subtracting thick and medium lines image from graphic image give the thin lines images. Fig. 1 show the thick, medium, and thin line image extracted from graphics image. This fine segmentation of walls is very useful especially when it is required to save floor plan in CAD format or to do 3D reconstruction of the building. Thick lines can be used to construct the boundary of the building (see Section 3.1.2) or 3D reconstruction of the external view of building. Medium lines represents internal structure and partition of the building and can be used to reconstruct the internal structure of the building. Thin lines represent different building elements, e.g., doors, windows, sofas, etc. However, the overall building structure is represented by the walls, therefore thick and medium lines are grouped together to get the walls image (see Fig. 1.6).

### 3.1.2. Structural analysis

The aim of structural analysis is to extract as much structural information as possible using previously segmented information. Structural analysis begins with the detection of the walls from the wall image as mentioned above. To vectorize the wall image the following approach is used, i.e., first, contours of the wall image are extracted using the method proposed by Suzuki and Abe (1985). Second, a polygonal approximation is performed on the extracted contours using simple chaining. It is based on compressing horizontal, vertical, and diagonal lines from detected contours so that only endpoints of these lines are left, which results in the vectorial representation for each contour (where a contour correspond to a polygon). These points are later considered as corner points of the approximated polygon for each contour. Finally, the polygon is constructed for each wall using these corner points. Each polygon represents a wall in the wall image.

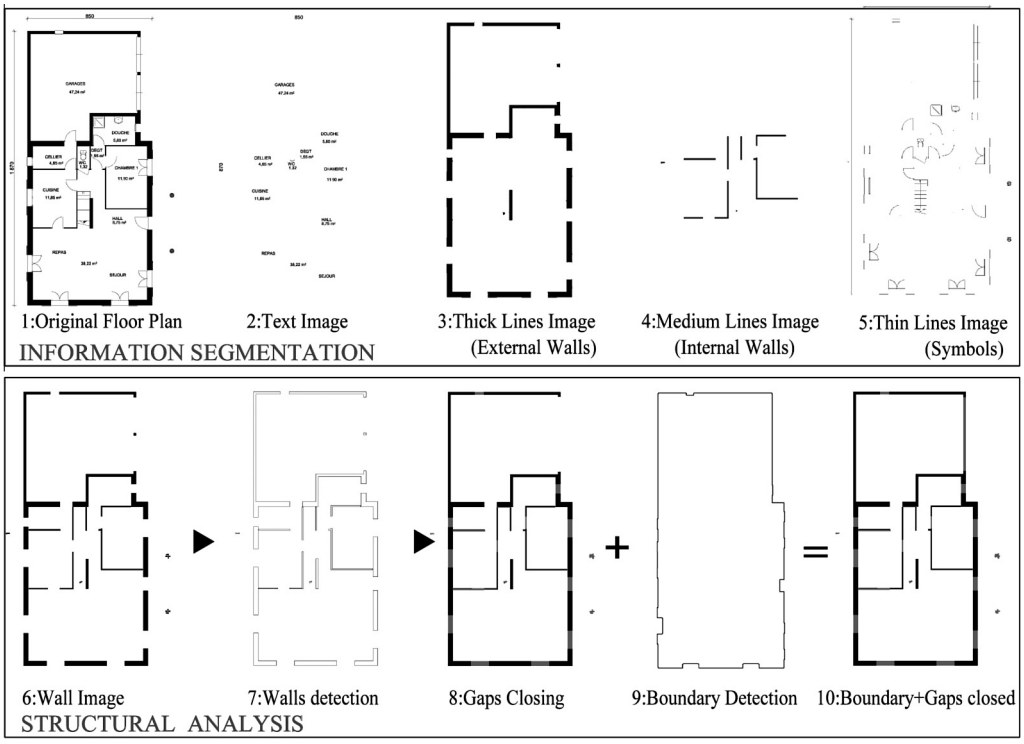



图1. 信息分割与结构分析。

彼此之间相互传递信息（见第3.1.1节）。其次，对所提取的信息结构进行分析，以获取房间的结构信息（见第3.1.2节）。最后，通过语义分析分别获取各房间的功能信息（见第3.1.3节）。

### 3.1.1. 信息分割

平面图包含的信息共同帮助建筑师表达建筑的实际空间布局与功能关系。在平面图分析过程中，不同类型的信息需要在不同的阶段加以解读。基于分而治之的策略，系统首先对信息进行分割处理。所提出方法的一个关键点在于对平面图中各类信息（如墙体、符号、文字等）进行精细化分割。这是必要的，因为对于某一特定步骤而言，不必要的信息只会构成噪声，并可能导致错误的结果。

首先，采用Ahmed等（2011b）提出的方法进行文本与图形的分割。该方法以Tombre等（2002）的算法为基础，并针对平面图进行了若干改进。文本/图形分割将平面图中的文本与图形分离。随后，文本图像（见图1.2）被用于语义分析，以提取房间功能信息。

图形图像进一步被分割为墙体以及其他建筑构件，如门、窗等。Dosch等人（2000）和Macé等人（2010）采用了一种粗线/细线分离算法，用于将墙体与符号区分开来。该算法将图像分解为两张子图像：一张包含墙体（包括外墙和内墙）的粗线图像，另一张包含符号的细线图像。我们对该方法进行了改进，增加了一类第三种线条，即中等粗细的线条。在所提出的算法中，通过依次执行腐蚀运算后再进行膨胀运算，即可分别获得粗线图像（表示外墙）、中等粗细线图像（表示内墙）以及细线图像（表示符号）。为了得到粗线图像，该过程需重复执行n次。

仅保留厚壁部分，其中 n 是根据训练集中的外墙厚度通过经验公式计算得到的。

随后，将粗线图像从图形图像中减去，得到包含中等和细线条的中间图像。在该结果图像上仅进行一次处理，即可获得中等线条图像。最后，从原始图形图像中分别减去粗线和中等线条图像，即可得到细线条图像。图1展示了从图形图像中提取的粗线、中等线条和细线条图像。这种精细的墙体分割在需要以CAD格式保存平面图或进行建筑物三维重建时尤为有用。粗线条可用于构建建筑边界（参见第3.1.2节）或进行建筑外立面的三维重建；中等线条代表建筑的内部结构与隔断，可用于重建建筑的内部结构；而细线条则表示各类建筑构件，如门、窗、沙发等。然而，建筑的整体结构主要由墙体体现，因此将粗线和中等线条合并，即可得到墙体图像（参见图1.6）。

### 3.1.2. 结构分析

结构分析的目的是利用先前分割得到的信息尽可能多地提取结构信息。结构分析首先从上述墙图像中检测墙体。为对墙图像进行矢量化，采用如下方法：首先，利用Suzuki和Abe（1985）提出的方法提取墙图像的轮廓；其次，对提取的轮廓采用简单链码法进行多边形逼近。该方法通过压缩检测到的轮廓中的水平、垂直及斜向线段，仅保留这些线段的端点，从而为每个轮廓生成矢量表示（其中每个轮廓对应一个多边形）。随后，将这些点视为各轮廓所对应近似多边形的角点；最后，基于这些角点为每面墙体构建多边形。每个多边形即代表墙图像中的一面墙体。

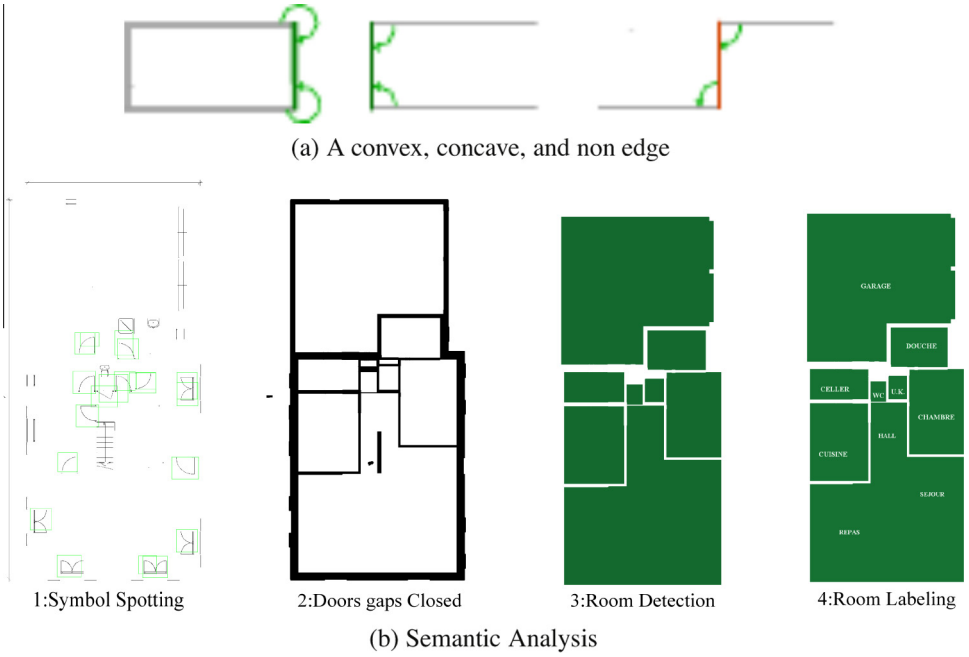



**Fig. 2.** Convex-Concave hypothesis, and Semantic Analysis.

The wall edges are then extracted from the detected walls to close the gaps between the walls. These gaps occur at elements like doors, windows, or sometime at gates. The process extracts all edges where those elements are likely to be found. To extract these edges we introduced convex/concave hypothesis. According to this hypothesis, a line of polygon is selected as wall edge if it is short and is either convex or concave (see Fig. 2a).

$$Walledge = \begin{cases} Convex & both\ angles\ are > 90^{\circ} \\ Concave & both\ angles\ are < 90^{\circ} \end{cases}$$

Fig. 1.7 shows the extracted wall edges from the walls image according to the convex/concave hypothesis. As a next step the gaps between the extracted edges are closed. Note, that here focus is to close only those gaps where windows or doors are likely to be found based on empirically defined thresholds $T_{merge}$. $T_{merge}$ is selected based on the size of doors and windows which are inside the building and is optimized on validation set. However, gaps at the outer walls are often larger than gaps occurring at doors inside the building. In order to merge even those larger gaps we compute the outer wall image and use boundary image of the building.

Extraction of the building boundary is another key point of the proposed approach. The extracted building boundary is used to close the large gaps which are normally due to gates and can be used to get the external structure of a building.

To extract the building boundary a convex hull of the wall image is created, and the portion of the floor plan image which is inside the convex hull is extracted, neglecting everything outside. Horizontal and vertical smearing is performed on extracted image to fill the gaps between the lines corresponding to windows and gates. To remove all the lines which are not part of the building structure (often they correspond to measurements), erosion and dilation is performed on the smeared image. After removal of these lines we can directly extract the external contours of the image. These contours approximate the building boundary, i.e., the external walls. In our experiments described in Section 4.1 we show the influence of this particular processing step.

*3.1.3. Semantic analysis*

The aim of semantic analysis is to extract the semantic information of the floor plan. While it is easy for a human to gather this information, its automation involves a high complexity. Semantic analysis spots different building elements in the floor plan and interprets them with respect to their context. In this paper we use the speeded up robust features (SURF) (Bay et al., 2008), which is a robust, translation, rotation, and scale invariant representation method. It first extracts the key points/points of interest from the image. Then each key point is represented by a discriminative descriptor. A standard door image serves as a reference template for SURF. Mainly arc is detected by SURF, therefore it able to detect both left and right doors.

Note that some erroneous symbols have been extracted by our approach. At a later step these symbol positions are matched with the gaps found during wall edge detection. Only those results which overlap with gates are taken into account as actual doors. Fig. 2b.2 shows the image where the gaps at the doors are closed.

To detect the actual bounds of rooms, the image with the closed gaps is inverted and connected component analysis is performed on it. All of the very small connected components are removed, where as each of the remaining connected component is referred as room. The detected rooms can be found in Fig. 2b.3. After detection of rooms the next step is to define there functions like WC, Living room etc. In order to find the function of each room, the text layer from the information segmentation as well as the connected component of the room is used. In particular, all text components which lie in the boundary of a room are taken into account. After extraction of the room text, horizontal and vertical smearing is performed on the extracted text to merge the neighboring characters, resulting in the bounds for words. Using the bounding boxes all the words are rotated to a horizontal direction and OCR[3] is performed on them. The OCR result is then compared to rooms title dictionary and the closest title according to the Levenshtein distance is assigned to the room. Note that before applying the dictionary, all the digits and special characters are removed from the OCR result. The rooms which do not have any physical partition contain more than one label.

[3] Tessaract has been used for performing OCR.

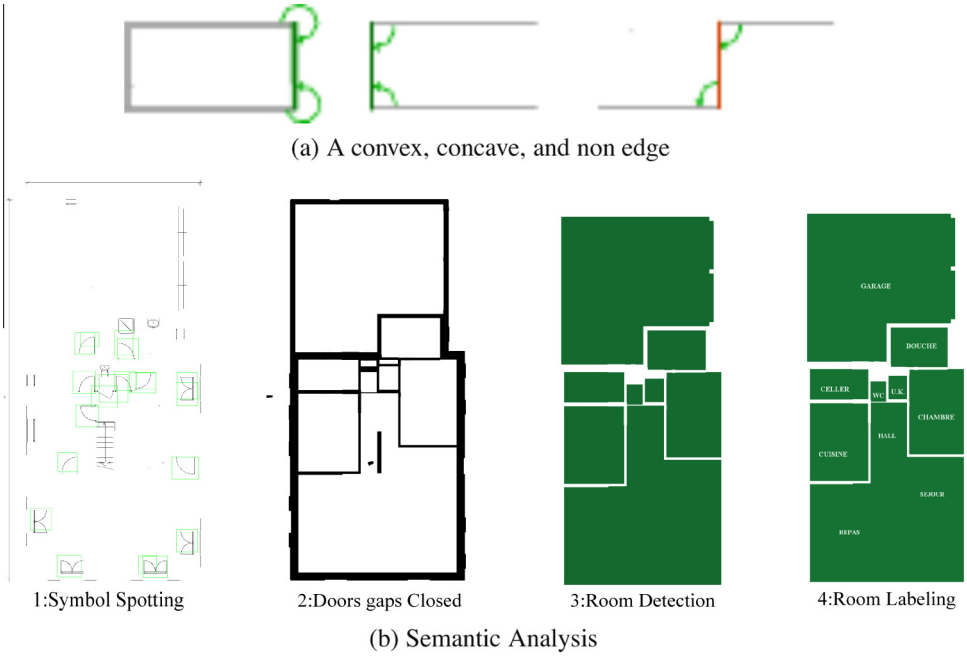



图2. 凸-凹假说与语义分析。

随后，从检测到的墙体中提取墙体边缘，以封闭墙体之间的缝隙。这些缝隙通常出现在门、窗或某些情况下还可能出现在大门等构件处。该过程会提取所有可能存在这些构件的边缘。为提取这些边缘，我们提出了凸/凹假设：若多边形的一条边较短且呈凸形或凹形，则将其判定为墙体边缘（见图2a）。

3.1.3 语义分析

语义分析的目的是提取平面图中的语义信息。尽管人类能够较为容易地获取这些信息，但其自动化实现却具有较高的复杂性。语义分析能够在平面图中识别出不同的建筑构件，并结合其上下文对其进行解释。在本文中，我们采用加速稳健特征（SURF）算法（Bay等，2008），这是一种具有平移、旋转和尺度不变性的鲁棒表示方法。该算法首先从图像中提取关键点或感兴趣点，然后为每个关键点构建一个具有判别能力的描述子。以标准门图像作为SURF的参考模板。由于SURF主要检测弧线特征，因此能够同时识别左开门和右开门。

$$\text{Walledge} = \begin{cases} \text{Convex} & \text{both angles are} > 90^{\circ} \\ \text{Concave} & \text{both angles are} < 90^{\circ} \end{cases}$$

图1.7展示了根据凸/凹假设从墙体图像中提取的墙体边缘。下一步是对所提取边缘之间的缝隙进行闭合处理。需要注意的是，此处仅对那些基于经验确定的阈值$T_{merge}$而可能含有门窗的缝隙进行闭合。阈值$T_{merge}$则依据建筑物内部门窗的尺寸设定，并在验证集上进行优化。然而，外墙处的缝隙往往大于建筑物内部门处的缝隙。为了连通甚至这些较大的缝隙，我们首先计算外墙图像，并利用建筑物的边界图像。

建筑物边界提取是所提出方法的另一个关键环节。提取出的建筑物边界可用于闭合通常由门洞造成的较大空隙，并用于获取建筑物的外部轮廓结构。

为了提取建筑物边界，首先对墙体图像构建凸包，并提取位于该凸包内部的楼层平面图像区域，忽略凸包外部的所有内容。随后对提取的图像进行水平和垂直方向的平滑处理，以填补与窗户及门对应的线条之间的间隙。为进一步去除不属于建筑结构的线条（通常对应于测量线），在平滑后的图像上依次执行腐蚀与膨胀操作。完成这些线条的去除后，即可直接提取图像的外轮廓，这些轮廓近似地反映了建筑物的边界，即外墙的位置。在第4.1节所描述的实验中，我们将展示这一特定处理步骤的影响。

需要注意的是，我们的方法提取出了部分错误的符号。在后续步骤中，这些符号的位置将与墙体边缘检测过程中发现的缺口进行匹配。只有与门洞重合的结果才会被认定为实际的门。图2b.2展示了门洞处的缺口已被闭合后的图像。

为检测房间的实际边界，对已闭合缝隙的图像进行取反操作，并在其上执行连通分量分析。剔除所有极小的连通分量，剩余的每个连通分量即被视为一个房间。所检测到的房间如图2b.3所示。在房间检测完成后，下一步是确定各房间的功能，例如卫生间、客厅等。为了识别每个房间的功能，同时利用信息分割得到的文本层以及该房间的连通分量。具体而言，将位于房间边界内的所有文本组件纳入考量。提取房间内的文本后，对其分别进行水平和垂直方向的平滑处理，以合并相邻字符，从而获得单词的边界框。基于这些边界框，将所有单词旋转至水平方向，并对其执行OCR（版本v4）处理。随后，将OCR结果与房间标题词典进行比对，根据Levenshtein距离选取最接近的标题作为该房间的标签。需要注意的是，在应用词典之前，须从OCR结果中去除所有数字及特殊字符。对于未设置物理隔断的房间，其标签可能包含多个类别。

[3] Tessaract has been used for performing OCR.

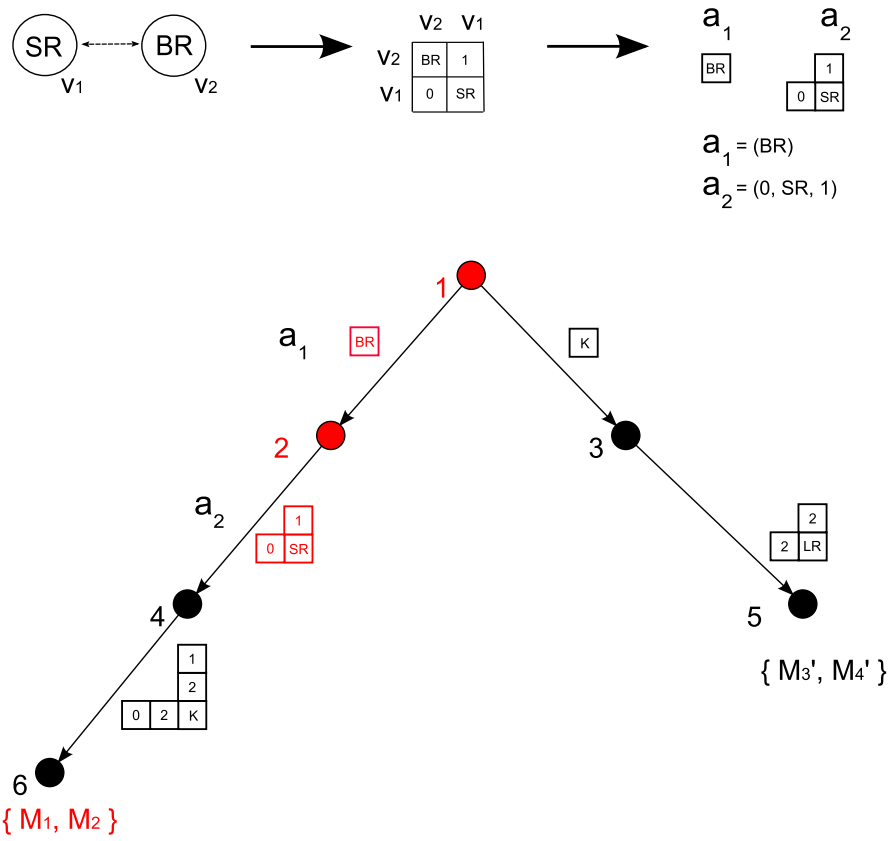



**Fig. 3.** The query graph with its two vertices $v_1$ and $v_2$ is represented using an adjacency matrix. The ordering of the vertices has to fulfill the well-founded order. The matrix is the split into its row column vectors $a_1, a_2$. These vectors will be used to find the path in the decision tree which is used for retrieval.

### 3.2. Sketch-based retrieval

As we are dealing with visual information, and an exclusive textual description of floor plans is too fuzzy, we propose a simplified visual query language. Whenever the architect is searching the repository, he formalizes his query as a sketch, similar to the fundamental concepts of Spatial-Query-by-Sketch proposed by Egenhofer (1996). Initially, the architect sketches a floor plan with the associated rooms, zones and units (see Fig. 6b). The respective online data of the pen device used to detect the geometrical shapes (see Fig. 6c) representing the concepts and gestures, which indicate the connection type.

Sketch recognition is performed using the following procedure. During sketching the online pen data is cached. As soon as the user stops drawing for a certain period of time, the recognition of the previous components starts. For the shape detection we used the Vision Objects shape detection.[4]

Currently we use the following visual query language:

- **Rectangles** represent **structural entities**,
- **Enclosings** imply **part-of relation**,
- **Single lines** indicate **adjacent connections**,
- **Two parallel lines** indicate **direct connections**.

The schematic abstraction is now interpreted and translated into the proposed semantic structure by Langenhan and Petzold (2010). Finally, when the user triggers the search, the scene which is composed of rectangles and lines will be analyzed. The type of the drawn entity or connection currently is defined by the user by using handwritten annotations on the rectangles. Further work is to interpret the natural symbolism of architects instead of the simplified query language.

### 3.3. Graph structure

In this section, we discuss how the extracted graph structure can be used for the retrieval. We propose our concept for the retrieval. The extracted semantics are represented as a graph $G = (V, E)$. The vertices $V$ have a type $T_{vertex}$ reflecting a level, unit, zone or room. The edges $E$ also have different types $T_{edge}$ indicating if the vertices are connected directly or are just adjacent, both of these relations are symmetric. The *part_of* relation indicates which vertex adheres to a vertex of a superior type $T_{vertex}$, for instance a sleeping room which is part of a sleeping zone.

The retrieval algorithm is based on a modified version of Messmer and Bunke (1999) algorithm, where the row column vectors of the adjacency matrices are arranged in a decision tree. The modified method Weber et al. (2011) introduces a well-founded total order on the graph labels. In the original method Messmer and Bunke (1999) all permutations of the adjacency matrix, representing the graph, have to be determined. Whereas for the proposed method the permutations which violate the introduced order are not allowed. Hence the number of possible permutations is limited. The row-column vectors for each permutation of the adjacency matrix are used to compile the decision tree. The decision tree contains multiple graphs and acts as an index for real-time sub-graph matching.

So, during run-time the decision tree is loaded into memory and by traversing the decision tree, the corresponding subgraph matri-

[4] http://www.visionobjects.com/: Last accessed 10/25/2011.

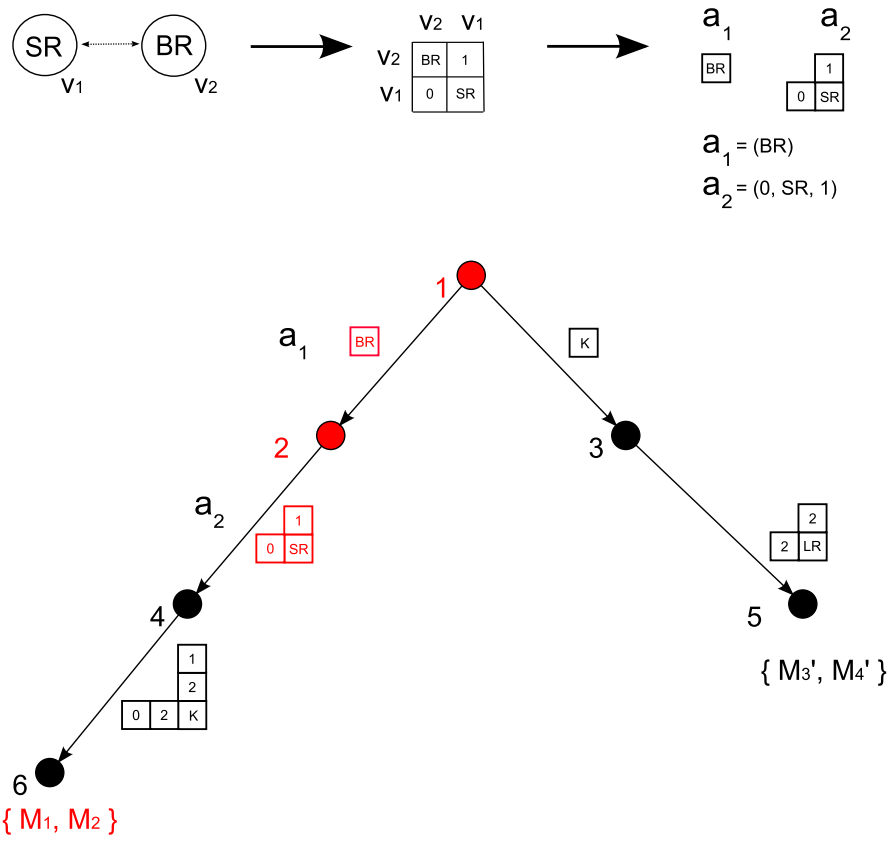



图 3. 包含两个顶点 $v_1$ and 和 $v_2$ is represented using an adjacency matrix. The ordering of the vertices has to fulfill the well-founded order. The m 的查询图 矩阵是t 它被分解为其行向量和列向量 $a_1$ 和 $a_2$。这些向量将用于确定决策树中用于检索的路径。

### 3.2. 基于草图的检索

由于我们处理的是视觉信息，而仅依赖文字描述的平面图过于模糊，因此我们提出了一种简化的视觉查询语言。当建筑师在检索知识库时，他会将自己的查询形式化为一张草图，这与Egenhofer（1996）提出的“基于草图的空间查询”这一基本概念类似。起初，建筑师会绘制一张包含相关房间、区域和单元的平面图草图（见图6b）。随后，用于捕捉几何形状的笔式输入设备会实时采集相应的在线数据（见图6c），这些数据既表征了各概念，也标识了连接类型。

草图识别采用如下流程进行。在用户绘制过程中，实时笔迹数据会被缓存。一旦用户停止绘制达到预设时长，系统即开始对先前绘制的各组件进行识别。对于形状检测，我们采用了Vision Objects提供的形状检测算法。[4]

目前我们使用以下可视化查询语言：

- 矩形表示结构实体，
- 包围关系表示“部分—整体”关系，
- 单线表示邻接连接，
- 双平行线表示直接连接。

该示意图抽象现已被Langenhan和Petzold（2010）解释并转化为所提出的语义结构。最后，当用户触发搜索时，由矩形和线条构成的场景将被分析。其类型为……

[4] http://www.visionobjects.com/: Last accessed 10/25/2011.

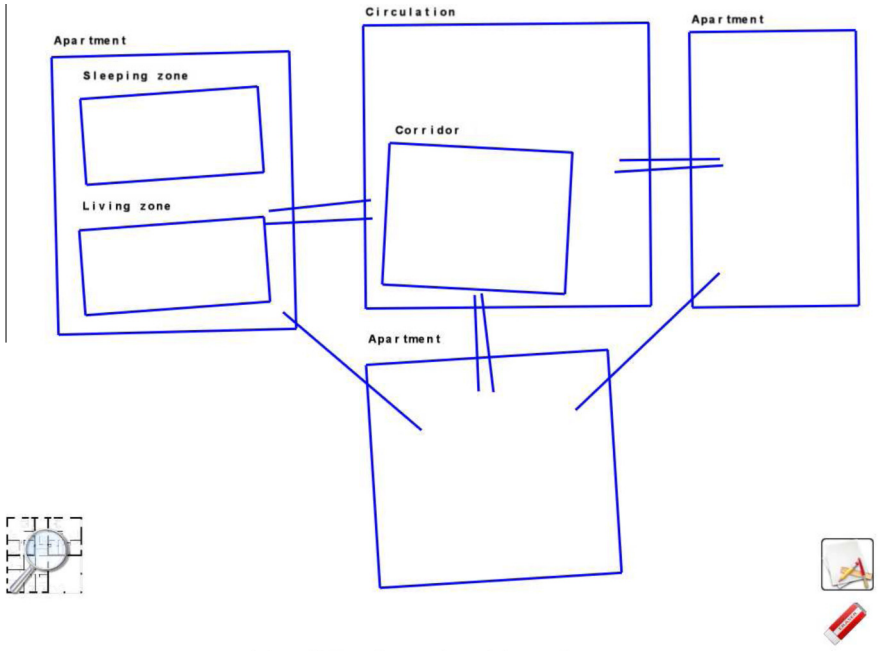
当前，所绘制的实体或连接由用户通过在矩形上添加手写标注来定义。后续工作将致力于解析建筑师的自然符号表达，而非依赖简化的查询语言。

### 3.3. 图结构

在本节中，我们讨论如何将提取的图结构用于检索。我们提出了相应的检索框架。所提取的语义以图G = (V, E)的形式表示，其中顶点集V具有类型$Tv_{ertex}$，用于表征层级、单元、区域或房间；边集E亦具有不同的类型$T_{edge}$，用以区分顶点之间是直接相连还是仅相邻，这两种关系均为对称关系。此外，部分—整体关系用于刻画某一顶点隶属于更高一级类型$Tv_{ertex}$的顶点，例如，作为睡眠区一部分的卧室。

该检索算法基于Messmer和Bunke（1999）算法的改进版本，其中邻接矩阵的行向量和列向量被组织成一棵决策树。Weber等人（2011）提出的改进方法在图标签上引入了一种有理论依据的全序关系。在原始方法中，必须枚举表示该图的所有邻接矩阵的排列；而在所提出的方法中，不允许出现违反该序关系的排列，从而有效减少了可能的排列数。针对邻接矩阵的每一种排列，其对应的行向量和列向量被用于构建决策树。该决策树包含多个图，并用作实时子图匹配的索引结构。

因此，在运行时，决策树被加载到内存中，并通过遍历该决策树来确定相应的子图矩阵——




(a) a.SCatch retrieval interface.

**Fig. 4.** a.SCatch retrieval interface.

**Table 2**
Room Detection results.

| | Macé et al. (2010) | Proposed | No boundary det. |
|---|---|---|---|
| Detection rate (%) | 85 | 94.88 | 77.47 |
| Rec. accuracy (%) | 69 | 81.3 | 86.88 |
| One to many count | 2 | 1.48 | 1.01 |
| Many to one count | 0.76 | 2.14 | 1.58 |

ces are classified. For the query graph $Q$ the adjacency matrix $M$ is determined following the constraints defined by ordering. Afterwards the adjacency matrix is split up into row-column vectors $a_i$. For each level $i$ of the decision tree the corresponding row-column vector $a_i$ is used to find the next node in the decision tree using an index structure. Fig. 3 provides an illustration of a sample query, where a sleeping room (SR) should be directly connected to a bath room (BR). Further details are presented in Weber et al. (2011) as well as experiments on graphs with up to 30 vertices.

Our current work focuses on researching approximative approaches for solving the subgraph matching problem. Furthermore, we are researching techniques to cluster the graph in our repository, by comparing floor plan graphs among each other and group similar ones together. Thus, if a query is not similar to a graph of a cluster it might also be not similar to the other graphs in the cluster.

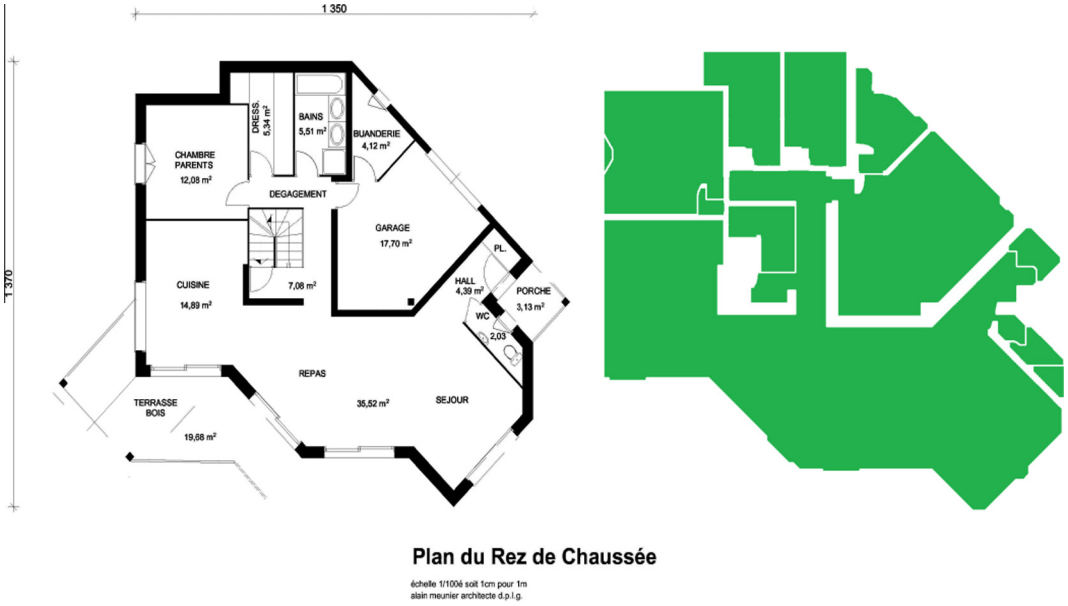
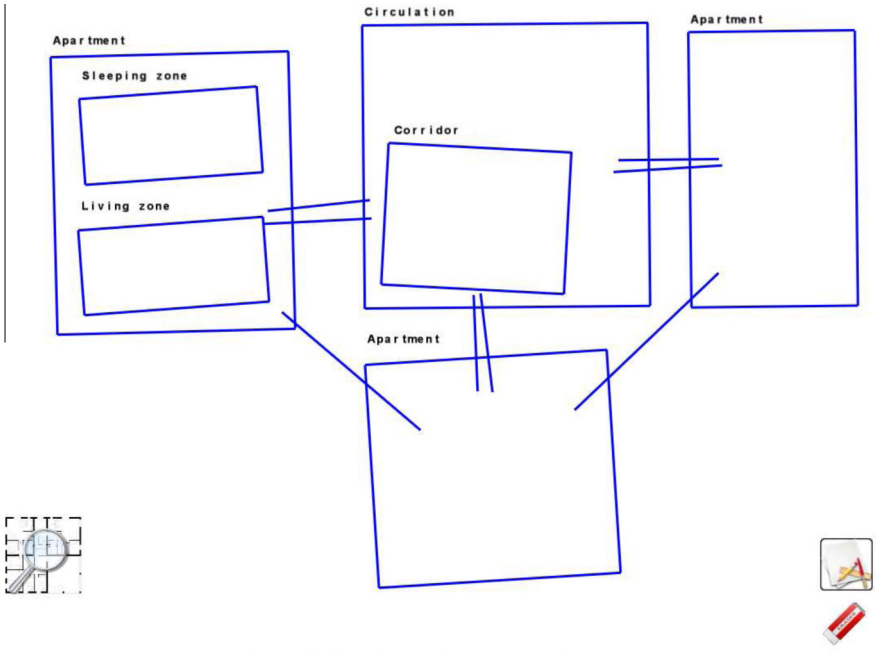



**Fig. 5.** Room detection result.




(a) a.SCatch retrieval interface.

图4. a. SCatch 检索界面。

表2 房间检测结果。

| | Macé et al. (2010) | Proposed | No boundary det. |
|---|---|---|---|
| Detection rate (%) | 85 | 94.88 | 77.47 |
| Rec. accuracy (%) | 69 | 81.3 | 86.88 |
| One to many count | 2 | 1.48 | 1.01 |
| Many to one count | 0.76 | 2.14 | 1.58 |

节点被分类。对于查询图Q，根据排序所定义的约束条件确定其邻接矩阵M。随后，该邻接矩阵被分解为行向量和列向量。

ai. 对于决策树的每一层 i，均使用相应的行-列向量 $a_i$，借助索引结构来确定决策树中的下一个节点。图3展示了一个示例查询，其中卧室（SR）应与浴室（BR）直接相连。更多细节见Weber等（2011），并附有针对最多包含30个顶点的图的实验结果。

我们目前的工作重点是研究用于求解子图匹配问题的近似方法。此外，我们还在探索一种技术，通过比较存储库中的平面布局图并将其相似者归为同一簇，从而对图进行聚类。因此，如果某个查询与某一簇中的某张图不相似，则它也可能与该簇中的其他图不相似。

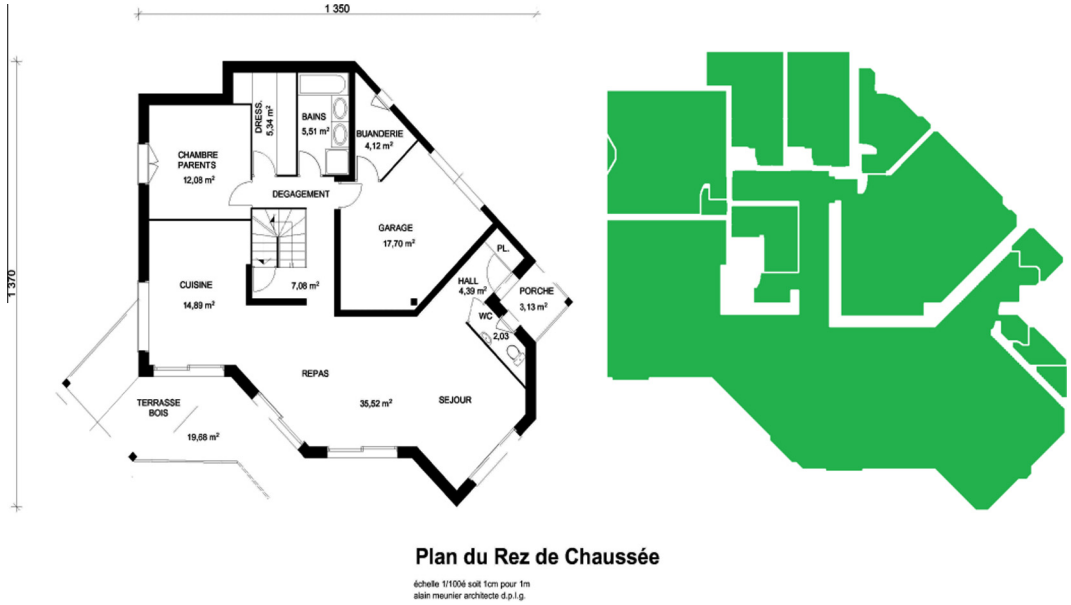



图5. 房间检测结果。

**Table 3**
Complexity and Detection rate for each query.

| Query | Quadrangles | | Adjacent C. | | Direct C. | |
|---|---|---|---|---|---|---|
| 1 | # | Corr | # | Corr | # | Corr |
| 1 | 8 | 0.96 | 2 | 0.95 | 2 | 0.8 |
| 2 | 5 | 0.96 | 2 | 09 | 3 | 0.9 |
| 3 | 8 | 0.93 | 2 | 0.95 | 5 | 0.7 |
| 4 | 6 | 0.95 | 3 | 0.93 | 1 | 0.7 |
| 5 | 3 | 1.0 | 2 | 1.0 | 1 | 1.0 |
| 6 | 9 | 0.99 | 3 | 0.97 | 5 | 0.96 |
| 7 | 6 | 0.98 | 2 | 0.95 | 4 | 0.65 |
| 8 | 9 | 0.99 | 4 | 0.93 | 4 | 0.9 |
| 9 | 10 | 0.97 | 1 | 0.8 | 4 | 0.9 |
| 10 | 3 | 0.97 | 1 | 0.9 | 2 | 0.9 |
| **Overall** | | 0.97 | | 0.93 | | 0.86 |

### 3.4. User interface

As the pen and touch paradigm is more intuitive for the work of an architect, the prototype is implemented for the *Touch* & *Write* table. The *Touch* & *Write* table (Liwicki et al., 2010) combines the paradigm of multi-touch input and pen input. Architects prefer to sketch in their initial design phase. A pen gives them more freedom than using a mouse with Computer Aided Design (CAD) software. Using the *Touch* & *Write* pen device to draw in a digital environment allows more immediate interaction and the architects immediately benefit from the digital representation of their drawings.

The pen is an adequate tool to sketch the current architectural problem. The results of the semantic search will be represented as graphical information and the touch interaction is an intuitive metaphor for interacting with the displayed information. For example, the architect is able interact with the graphical information using simple and intuitive gestures to zoom or navigate within the floor plan.

The a.SCatch system provides tools for two purposes. First, an input interface offers the architect a possibility to edit and correct the results of the automatic room and interconnection detection discussed in Section 3.1. Here the pen device is used to frame rooms, zones or units in a floor plan.

Second, the architect can sketch a query by using the discussed visual query language. The results are displayed as images ordered according to the calculated similarity measure. The interaction with the result is done by using touch gestures (see Fig. 4).

## 4. Experiments

### 4.1. Floorplan analysis evaluation

Our system is evaluated using a data set containing original floor plan images. This data set was introduced by Macé et al. (2010) and contains the floor plan images from the period of more the ten years. The size of each floor plan image in the data set is $2479 \times 3508$. All floor plans are binarized to ensure that only structural information of the floor plans is used for the analysis (and not the color information).

In order to report the accuracy of our system, we use the protocol introduced by Phillips et al. (1999). It allows reporting *exact match* (one to one) as well as *partial matches* (one to many and many to one). For further details refer to (Phillips et al., 1999).

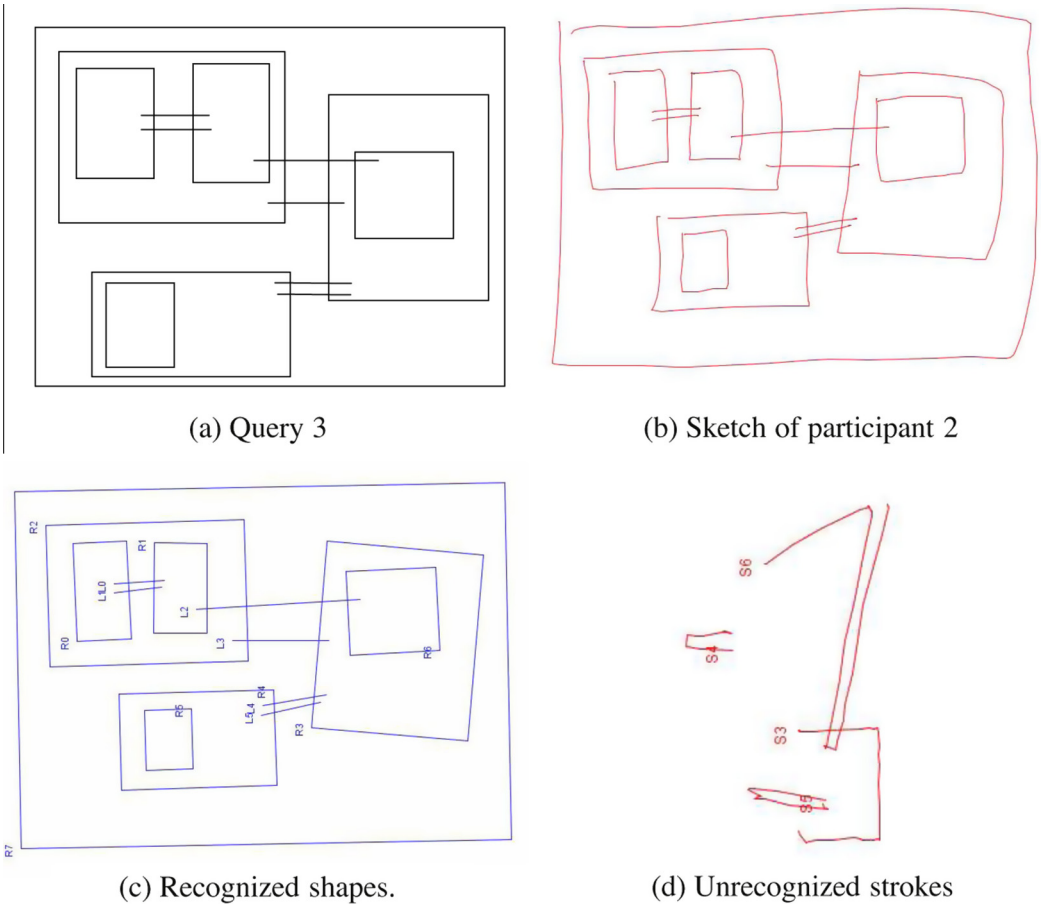



**Fig. 6.** Visualization of the results.

表3 各查询的复杂度与检测率。

| Query | Quadrangles | | Adjacent C. | | Direct C. | |
|---|---|---|---|---|---|---|
| 1 | # | Corr | # | Corr | # | Corr |
| 1 | 8 | 0.96 | 2 | 0.95 | 2 | 0.8 |
| 2 | 5 | 0.96 | 2 | 09 | 3 | 0.9 |
| 3 | 8 | 0.93 | 2 | 0.95 | 5 | 0.7 |
| 4 | 6 | 0.95 | 3 | 0.93 | 1 | 0.7 |
| 5 | 3 | 1.0 | 2 | 1.0 | 1 | 1.0 |
| 6 | 9 | 0.99 | 3 | 0.97 | 5 | 0.96 |
| 7 | 6 | 0.98 | 2 | 0.95 | 4 | 0.65 |
| 8 | 9 | 0.99 | 4 | 0.93 | 4 | 0.9 |
| 9 | 10 | 0.97 | 1 | 0.8 | 4 | 0.9 |
| 10 | 3 | 0.97 | 1 | 0.9 | 2 | 0.9 |
| **Overall** | | 0.97 | | 0.93 | | 0.86 |

### 3.4 用户界面

由于笔式输入与触控相结合的交互范式对于建筑师的工作而言更为直观，因此该原型被应用于“Touch & Write”桌面上。Touch & Write 桌面（Liwicki 等，2010）融合了多点触控输入与笔式输入的交互模式。在设计的初始阶段，建筑师更倾向于进行草图绘制；与使用鼠标配合计算机辅助设计（CAD）软件相比，笔式输入能够提供更大的操作自由度。借助 Touch & Write 笔式设备在数字化环境中进行绘图，可以实现更为即时的交互，使建筑师能够立即受益于其图纸的数字化表达。

笔是描绘当前建筑问题的合适工具。语义搜索的结果将以图形化信息的形式呈现，而触控交互则具有直观性。

用于与显示信息进行交互的隐喻。例如，建筑师可以通过简单直观的手势操作来缩放或在平面图中导航。

a.SCatch系统提供了两种用途的工具。首先，一个输入界面使建筑师能够对第3.1节中讨论的自动房间及连通性识别结果进行编辑与修正。在此过程中，通过使用绘图笔在平面图上框选房间、区域或单元。

其次，架构师可以利用所讨论的可视化查询语言绘制查询草图。查询结果以图像形式呈现，并按照计算得到的相似度度量进行排序。用户通过触控手势与结果进行交互（见图4）。

## 4. 实验

### 4.1. 平面布局分析评估

我们采用包含原始平面图图像的数据集对所提出的系统进行了评估。该数据集由Macé等人（2010）提出，其中包含了跨度超过十年的平面图图像。数据集中每幅平面图的尺寸为2479×3508像素。所有平面图均被二值化处理，以确保在分析过程中仅使用平面图的结构信息，而不引入颜色信息。

为了报告我们系统的准确性，我们采用了Phillips等人（1999）提出的协议。该协议既可用于报告完全匹配（一对一），也可用于报告部分匹配（一对多和多对一）。更多细节请参见（Phillips等，1999）。

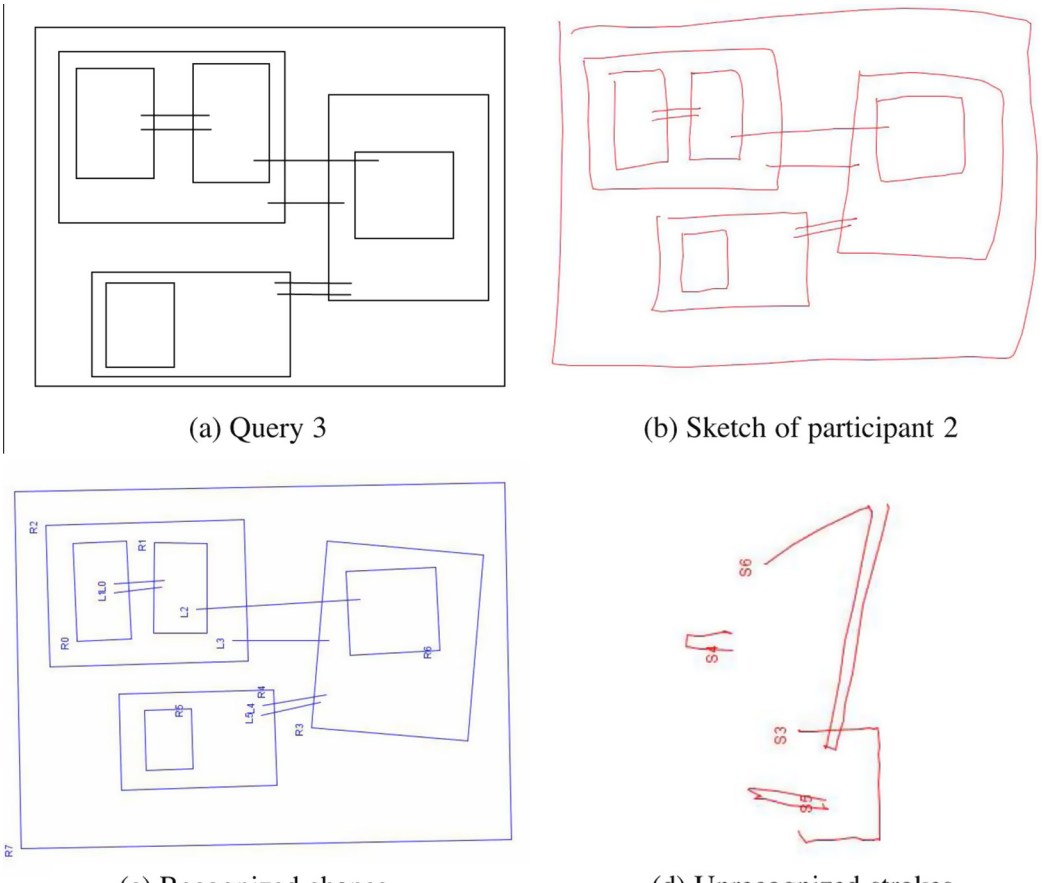



图6. 结果的可视化展示。

**Table 4**
Detection rate for each participant.

| Proband | Quadrangles | Adjacent C. | Direct C. |
|---|---|---|---|
| 1 | 1.0 | 1.0 | 0.97 |
| 2 | 0.96 | 0.82 | 0.81 |
| 3 | 0.93 | 0.82 | 0.77 |
| 4 | 0.91 | 0.95 | 0.74 |
| 5 | 0.99 | 1.0 | 0.87 |
| 6 | 1.0 | 0.9 | 0.97 |
| 7 | 0.97 | 0.95 | 0.83 |
| 8 | 0.97 | 0.95 | 0.84 |
| 9 | 0.99 | 0.95 | 0.94 |
| 10 | 0.97 | 0.91 | 0.84 |

Table 2 shows the results of rooms detection over the floor plan images dataset. The overall detection rate is 94.88%, which is approximately 10% higher than the 85% achieved in the reference system by Macé et al. (2010). More remarkably, the recognition accuracy has been improved by 12.30%. For around 20% of the images we received the recognition accuracy and detection rate both greater than 90%. In the worst case, the recognition accuracy and detection rate of our system were still 50% and 66.66% respectively. A further analysis shows the influence of the boundary detection (last column in Table 2) which was introduced in this paper. The detection rate is significantly improved.

The analysis of results in Table 2 reveals that our system has a good recognition accuracy and detection rate, along with less one to many count on average. This is because, a region is split into sub-regions only when a door or physical partition is found, in contrast to Macé et al. (2010) where regions are divided based on polygonal partitioning. To further reduce over segmentation, gap closing on the location where doors are detected need to be improved.

If there is no physical partition in the detected room no division is performed, which results in reduced segmentation overhead. Fig. 5 shows a very difficult example where large regions which do not contain any physical partition are marked as a single room. To further split these regions detailed semantic analysis can be done, in order to split large regions based on the measurement and text information available in the floor plan. This trend can also be seen in Table 2, where the *many to one count* is higher.

### 4.2. Retrieval evaluation

The query generation is a crucial part of the a.SCatch system, thus the first experiments focus on the visual language. We defined ten example queries covering different complexity levels (see Table 3) and and asked ten participants to copy these sketches, resulting in a total of 100 sketches. The participants were male and female students in the age between 23 and 29 years. All sketches were drawn on the *Touch* & *Write* and the handwritten strokes were recorded. To assess the pure recognition performance, we did not give a direct feedback of the recognized shapes. In order to measure the accuracy of the detection algorithm we count the correctly detected quadrangles and connections.

An example of a query taken for the evaluation is given in Fig. 6a. Furthermore Fig. 6b shows the recorded sketch of a participant. The detected shapes of the shape detection algorithm are illustrated in Fig. 6c.

For the evaluation we distinguished between detection rates for quadrangles, adjacent and direct connections. Table 3 shows the detection rate for each query. As can be seen, the detection rates for the quadrangles are very promising. Most of the other errors are due to missing quadrangles. Note that whenever a quadrangle is not detected all corresponding connections with this quadrangle will also be wrongly detected. Considering Query 2, 5, and 10, one can see that a high recognition rate of the quadrangles corresponds to a high recognition rate of the connections.

The detection rate for each participant is given in Table 4. An interesting observation is that there are persons who have a very nice way of drawing which is easily to be recognize. Other users tend to produce gaps in-between the strokes or draw very rough quadrangles, resulting in wrong recognition of the quadrangles and all corresponding connections, as illustrated in Fig. 6d.

As the interactive system acknowledges correct detected shapes, the user of the system has the chance to correct misinterpreted drawings before triggering a search. It is a good result that only one out of twenty rectangles needs to be corrected, making the sketch recognition already practically useful.

We have also performed experiments querying for reference floor plans using our graph search algorithm described in Section 3.3. The main complexity is the generation of the decision tree. Our optimization of the algorithm lead into a decrease of the tree nodes from $3.10 \times 10^{31}$ to $1.09 \times 10^{13}$. More details are reported in Weber et al. (2011).

## 5. Conclusion and future work

In this paper we have presented an intuitive system for searching floor plans by using a sketch-based interface. Furthermore, a complete system for automatic floor plan analysis is presented which is able to extract structural and semantic content of floor plan from given image. To represent the content of floor plan a graph-based semantic structure and associated visual query language is also presented in this paper. The recognition rates of sketch recognition as well as floor plan analysis are already good for the use in practice.

Our floor plan analysis system has been evaluated on a database from the literature. We outperform previous state-of-the-art methods and achieve a perfect recognition rate on several documents. Our experiments have shown that the proposed method works very well on a large corpus of 90 floor plans. However, in practice more different types of floor plans exist. We will adopt our methods to other types of plans in our future work.

To improve the detection rates of sketches, a dynamic programming approach to combine strokes (Feng et al., 2008) or using a combination of an online and offline detection can be used. Further work for the interactive system will be to offer the architect the freedom to choose between the proposed visual query language or let him sketch an initial floor plan with his symbolism and extract the semantic graph structure directly from this sketch.

Possible improvements for floor plan analysis consist of normalization of the contours and removing graphical elements outside the outer walls. Furthermore, a weak point of our approach is that it is only able to find the physical existent rooms. This means that if there is a large region and there is no wall with in this region, it will be marked as single room. However, architects tend to divide those rooms still into several functional rooms. This division can be achieved by a more sophisticated semantic analysis based on room labeling results.

In summary, the a.SCatch system is a successful approach integrating state-of-the-art offline image processing and online sketch recognition technologies. Together with the use of recent progress in knowledge management, it provides a novel powerful tool for sketch-based document work which can be applied to other application areas as well.

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表4 各参与者的检出率。

| Proband | Quadrangles | Adjacent C. | Direct C. |
|---|---|---|---|
| 1 | 1.0 | 1.0 | 0.97 |
| 2 | 0.96 | 0.82 | 0.81 |
| 3 | 0.93 | 0.82 | 0.77 |
| 4 | 0.91 | 0.95 | 0.74 |
| 5 | 0.99 | 1.0 | 0.87 |
| 6 | 1.0 | 0.9 | 0.97 |
| 7 | 0.97 | 0.95 | 0.83 |
| 8 | 0.97 | 0.95 | 0.84 |
| 9 | 0.99 | 0.95 | 0.94 |
| 10 | 0.97 | 0.91 | 0.84 |

表2展示了在楼层平面图数据集上进行房间检测的结果。总体检测率为94.88%，比Macé等人（2010）所提出的参考系统达到的85%高出约10个百分点。更为显著的是，识别准确率提升了12.30%。在约20%的图像中，识别准确率和检测率均超过90%。在最差的情况下，本系统的识别准确率和检测率仍分别达到50%和66.66%。进一步分析表明，本文引入的边界检测方法（表2最后一列）对检测效果产生了显著的提升。

表2中的结果分析表明，本系统具有较高的识别准确率和检测率，且平均的一对多匹配次数较少。这是因为在本系统中，仅当检测到门或实体隔断时才会将区域划分为子区域，而Macé等（2010）则基于多边形划分进行区域分割。为进一步减少过分割现象，还需改进在检测到门的位置处的缝隙填补算法。

如果在检测到的房间内不存在任何物理隔断，则不进行分割，从而降低分割开销。图5展示了一个非常复杂的示例，其中未包含任何物理隔断的大区域被标记为单个房间。为进一步划分这些区域，可基于楼层平面图中提供的测量数据和文本信息开展详细的语义分析。这一趋势在表2中亦有体现，其中“多对一”映射的数量较高。

### 4.2. 检索评估

查询生成是 a.SCatch 系统中的关键环节，因此首批实验聚焦于视觉语言。我们设计了十组示例查询，涵盖不同的复杂度级别（见表 3），并要求十名参与者临摹这些草图，最终共获得100份草图。参与者为年龄介于23至29岁之间的男女学生。所有草图均在 Touch & Write 设备上绘制，手写笔画被实时记录。为评估纯粹的识别性能，我们未对识别出的形状提供即时反馈。为了衡量检测算法的准确性，我们统计了正确检测到的四边形及其连接关系的数量。

用于评估的一个查询示例如图6a所示。此外，图6b展示了某位参与者的手绘草图。形状检测算法所识别出的形状如图6c所示。

在评估过程中，我们区分了对四边形、邻接连接以及直接连接的检测率。表3展示了各查询的检测率。可以看出，四边形的检测率非常理想；其余错误大多源于四边形的缺失。需要注意的是，只要某个四边形未被检测到，与其相关的所有连接也将被错误地检测。结合查询2、5和10来看，四边形的高识别率与连接的高识别率呈正相关。

每位参与者的识别率见表4。一个有趣的观察是，有些参与者绘图风格非常优美，易于识别；而另一些用户则倾向于在笔画之间出现断续，或绘制出较为粗糙的四边形，从而导致四边形及其所有相关连接的识别错误，如图6d所示。

随着交互系统对正确检测到的形状予以确认，用户在触发搜索之前有机会纠正被误识别的草图。令人欣慰的是，每二十个矩形中仅需更正一个，这使得草图识别技术已具备实际应用价值。

我们还利用第3.3节中所述的图搜索算法，开展了基于参考平面图的查询实验。其中的主要复杂性在于决策树的生成。通过对该算法的优化，决策树的节点数由3.10 × 10^v0降至1.09 × 10^v2；更多细节详见Weber等（2011）。

## 5. 结论与未来工作

本文提出了一种基于草图交互的直观平面图检索系统。此外，我们还构建了一个完整的平面图自动分析系统，能够从给定图像中提取平面图的结构与语义信息。为表征平面图的内容，本文进一步提出了基于图的语义结构及相应的视觉查询语言。目前，该系统的草图识别与平面图分析准确率已足以满足实际应用需求。

我们的楼层平面图分析系统已在文献中的数据库上进行了评估。在多项指标上，我们优于现有的最先进方法，并在若干文档上实现了完美的识别率。实验结果表明，所提出的算法在包含90张楼层平面图的大型语料库上表现优异。然而，在实际应用中，还存在更多不同类型的楼层平面图。在未来的工作中，我们将把该方法推广应用于其他类型的平面图。

为提高草图的识别率，可采用一种结合笔画的动态规划方法（Feng等，2008），或同时使用在线与离线识别技术。针对该交互式系统，后续工作将为建筑师提供自由选择权：既可使用所提出的可视化查询语言，也可允许其以自定义符号绘制初始平面布局，并直接从该草图中提取语义图结构。

针对平面图分析的潜在改进措施包括对轮廓线进行归一化处理，并去除外墙以外的图形元素。此外，本方法的一个不足之处在于，其仅能识别实际存在的房间。这意味着，若存在一个较大的区域且该区域内未设置墙体，则该区域将被标记为单一房间。然而，建筑师通常仍会将此类大空间划分为若干功能房间。这种划分可通过基于房间标注结果的更复杂的语义分析来实现。

总之，a.SCatch系统是一种成功的方法，它整合了最先进的离线图像处理技术和在线草图识别技术。结合近年来知识管理领域的最新进展，该系统为基于草图的文档处理提供了一种新颖而强大的工具，并且也可应用于其他应用领域。

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